

Duality and Simplex Algorithm

$$\min: c^T x$$

$$\text{st: } Ax = b$$

$$x \geq 0$$

$$\max: p^T b$$

$$\text{st: } p^T A \leq c^T$$

$$p \text{ free.}$$

Primal vs Dual Feasibility

-cost	$\bar{c} = c - c_B^T A_B^{-1} A$
$A_B^{-1} b$	$A_B^{-1} A$

$x_B \geq 0$ ← Primal simplex maintains primal feasibility
 works towards improving cost, more specifically,
 works towards: $0 \leq \bar{c} = c - \underbrace{c_B^T A_B^{-1}}_{p^T} A$, i.e., $p^T A \leq c$.

Dual simplex: Maintain dual feasibility: $p^T A \leq c$

and works towards primal feasibility: $A_B^{-1} b \geq 0$

Terminology: For B s.t. $A_B^{-1} b \not\geq 0$, $x = \begin{pmatrix} x_B \\ 0 \end{pmatrix}$ is called a Basic Solution, but not a BFS.

Dual Simplex Algorithm — 1/2

Have a BS : $B \subseteq [n]$, $|B| = m$,
but $x_B = A_B^{-1} b$ not comp. wise ≥ 0 .

	$\bar{c} = c - c_B^T A_B^{-1} A$
$A_B^{-1} b$	$A_B^{-1} A$

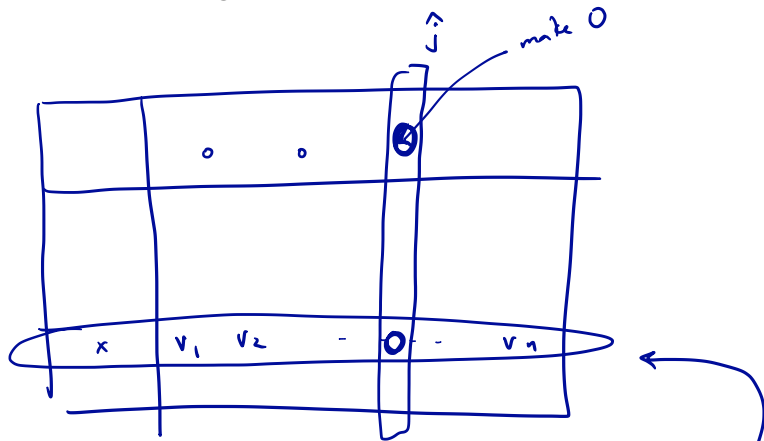
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Assume B is s.t.

$$\bar{c} \geq 0.$$

want $A_B^{-1} b \geq 0$.

Dual Simplex Algorithm — 2/2



Choose row: pick a row with $x_{B_i} < 0$ // pivot row

Pivot column: $\hat{j} : \min_{i: v_i < 0} \frac{\bar{c}_i}{|v_i|}$